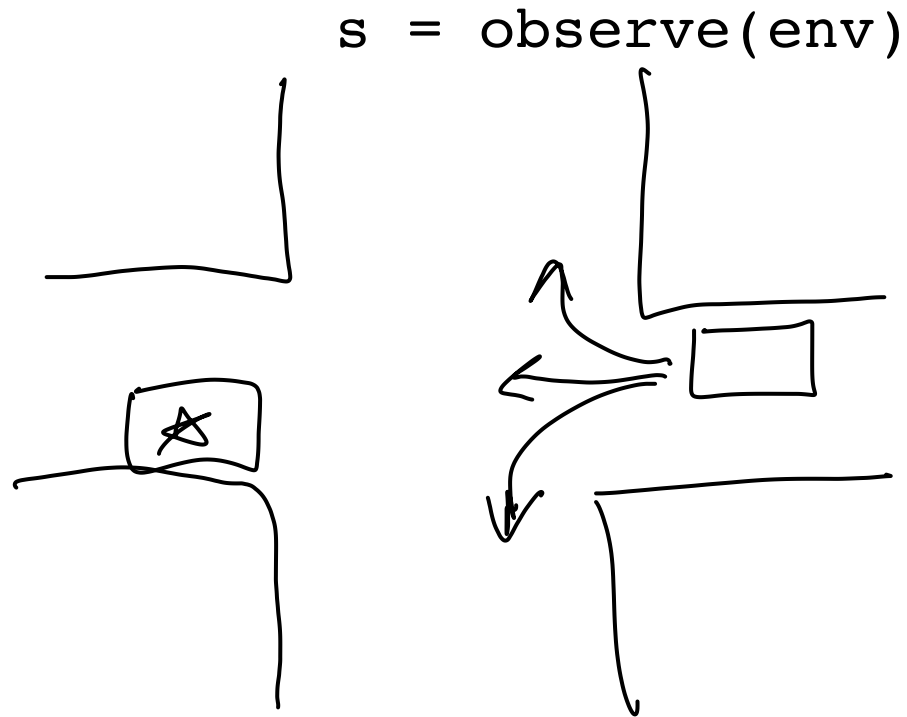


POMDPs

Ben Kraske

POMDPs

- We've been living a lie:



$$S = \{L\}$$

$$S = \{L, I\}$$

$$\mathcal{O} = \{I\}$$

Types of Uncertainty

Types of Uncertainty

Alleatory

Types of Uncertainty

Alleatory



Types of Uncertainty

Alleatory



Epistemic (Static)

Types of Uncertainty

Alleatory



Epistemic (Static)



.

Types of Uncertainty

Alleatory



Epistemic (Static)



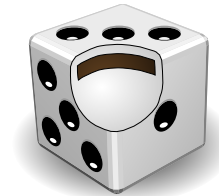
Epistemic (Dynamic)

Types of Uncertainty

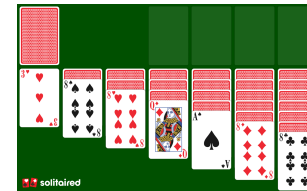
Alleatory



Epistemic (Static)



Epistemic (Dynamic)



Types of Uncertainty

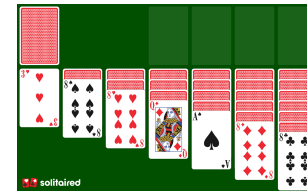
Alleatory



Epistemic (Static)



Epistemic (Dynamic)



Interaction

Types of Uncertainty

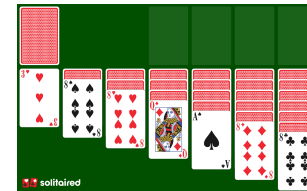
Alleatory



Epistemic (Static)



Epistemic (Dynamic)



Interaction

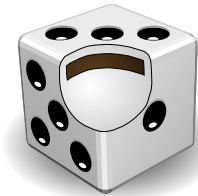


Types of Uncertainty

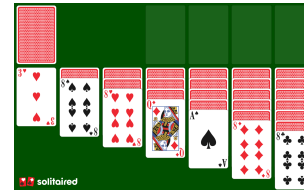
Alleatory



Epistemic (Static)



Epistemic (Dynamic)



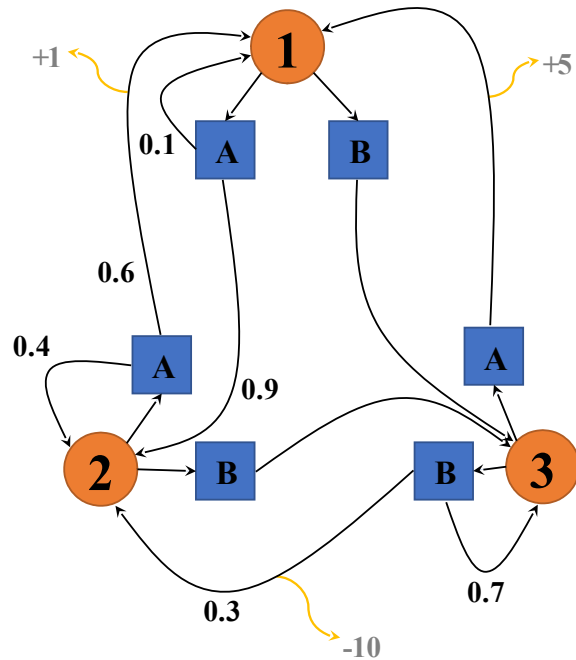
Interaction



POMDPs!

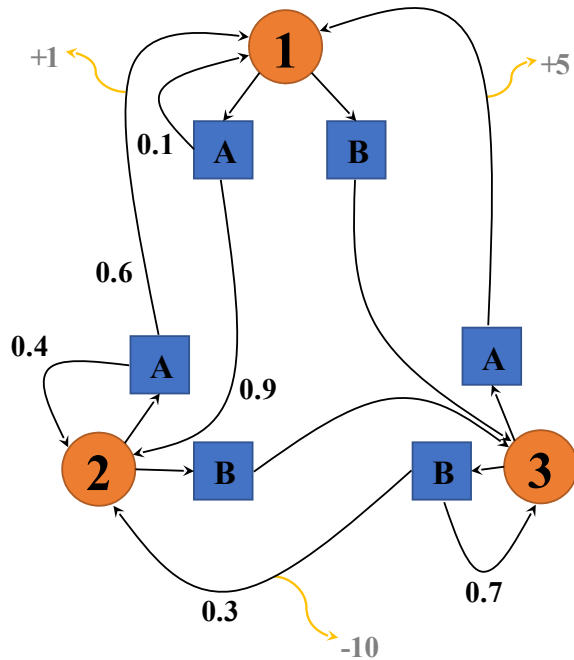
Markov Decision Process (MDP)

- $\mathcal{S}$ - State space
- $T(s' | s, a)$ - Transition probability distribution



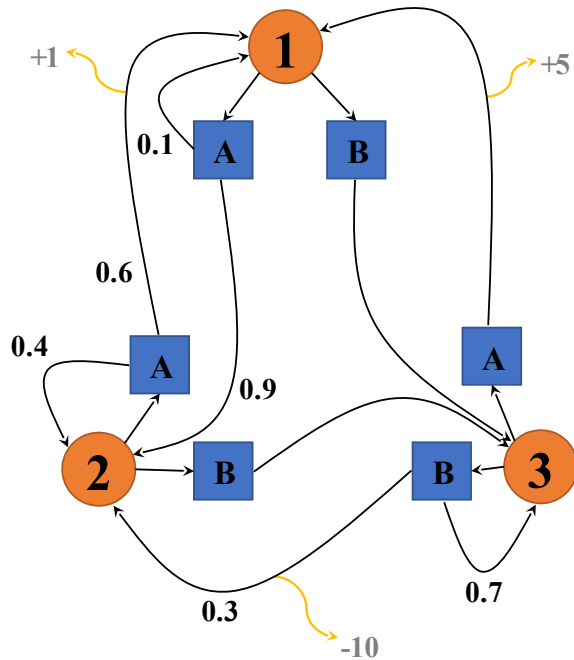
Markov Decision Process (MDP)

- $\mathcal{S}$ - State space
- $T(s' \mid s, a)$ - Transition probability distribution
- $\mathcal{A}$ - Action space



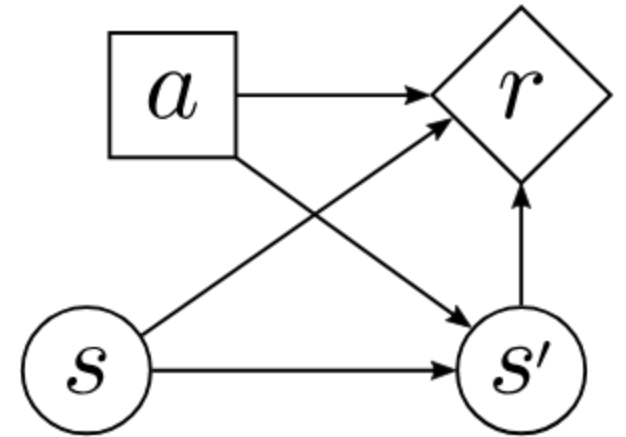
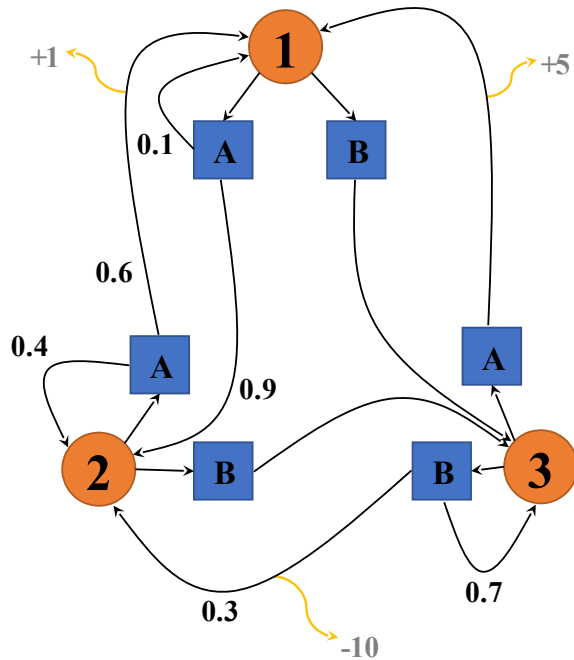
Markov Decision Process (MDP)

- $\mathcal{S}$ - State space
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- $\mathcal{A}$ - Action space
- $R(s, a)$ - Reward



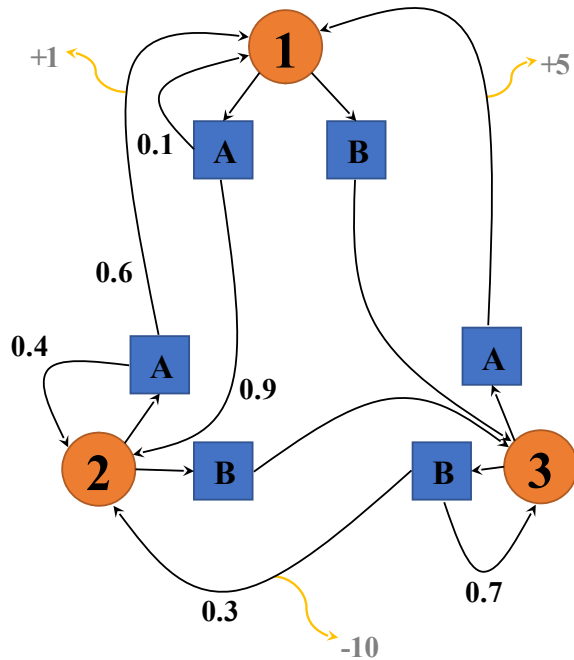
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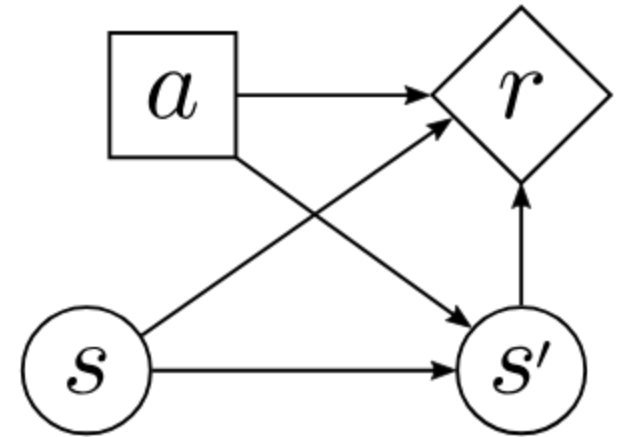


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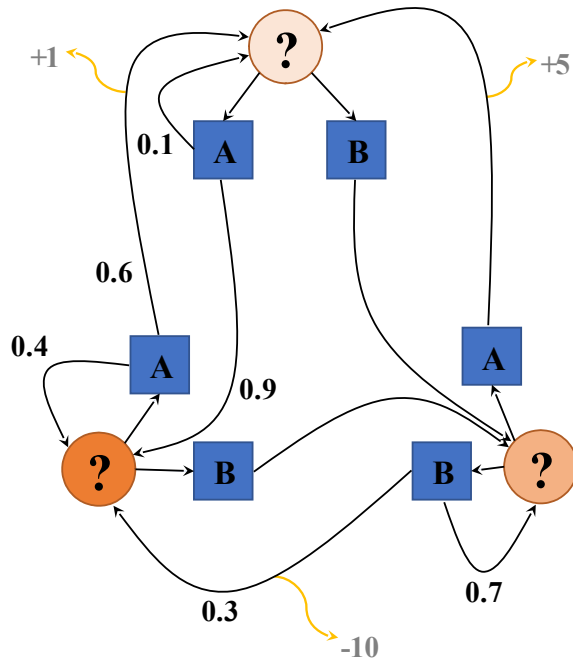


Alleatory



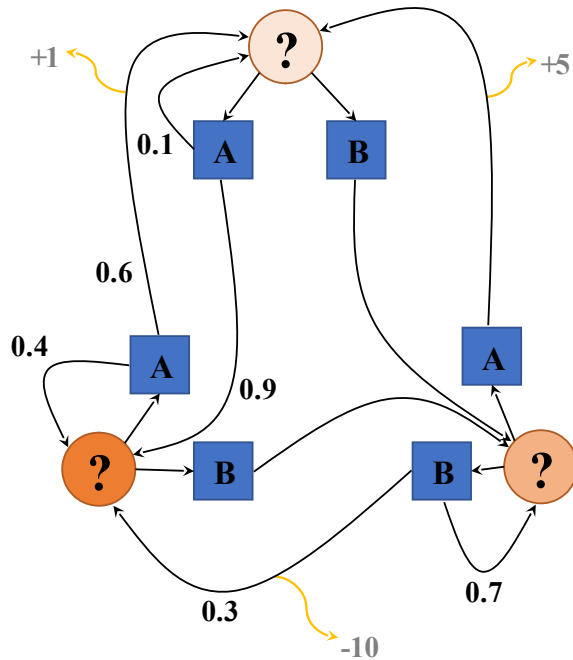
Partially Observable Markov Decision Process (POMDP)

- $\mathcal{S}$ - State space
- $T(s' | s, a)$ - Transition probability distribution
- $\mathcal{A}$ - Action space
- $R(s, a)$ - Reward



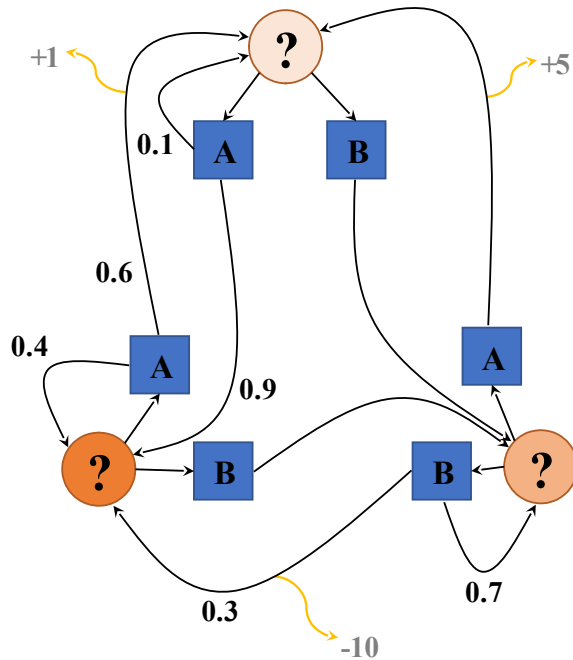
Partially Observable Markov Decision Process (POMDP)

- $\mathcal{S}$ - State space
- $T(s' \mid s, a)$ - Transition probability distribution
- $\mathcal{A}$ - Action space
- $R(s, a)$ - Reward
- $\mathcal{O}$ - Observation space

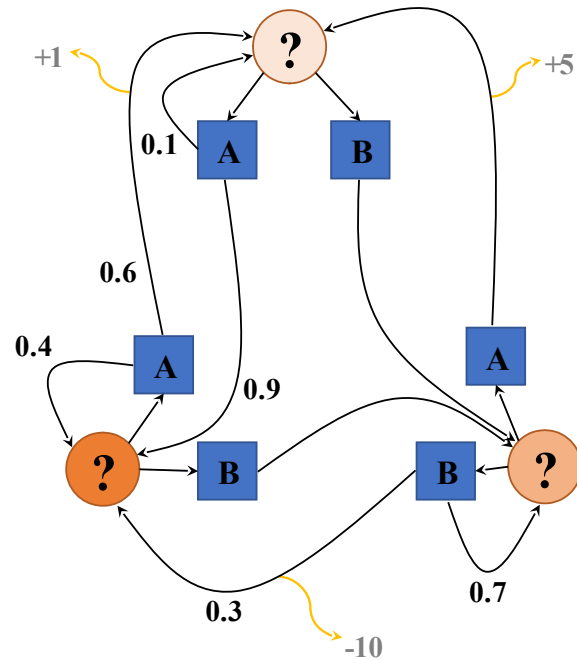


Partially Observable Markov Decision Process (POMDP)

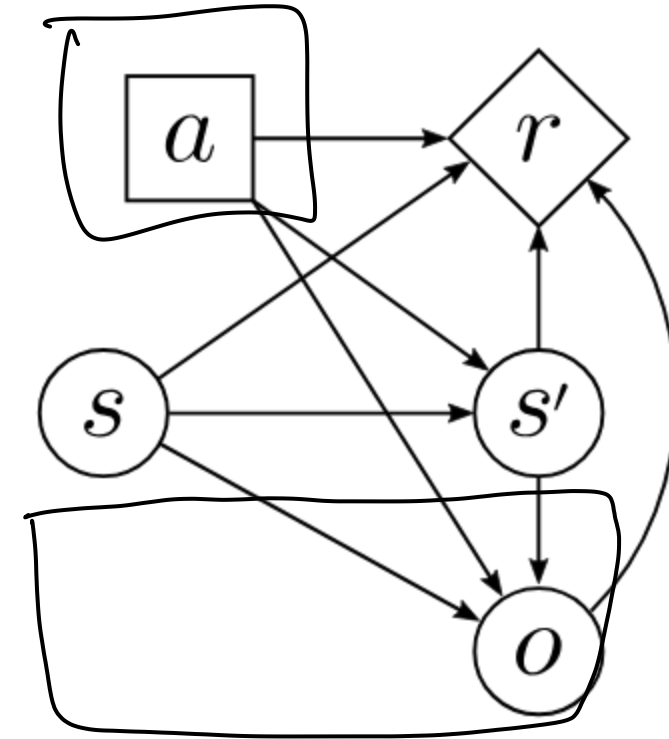
- $\mathcal{S}$ - State space
- $T(s' | s, a)$ - Transition probability distribution
- $\mathcal{A}$ - Action space
- $R(s, a)$ - Reward
- $\mathcal{O}$ - Observation space
- $Z(o | a, s')$ - Observation probability distribution



Partially Observable Markov Decision Process (POMDP)



- $\mathcal{S}$ - State space
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- $\mathcal{A}$ - Action space
- $R(s, a)$ - Reward
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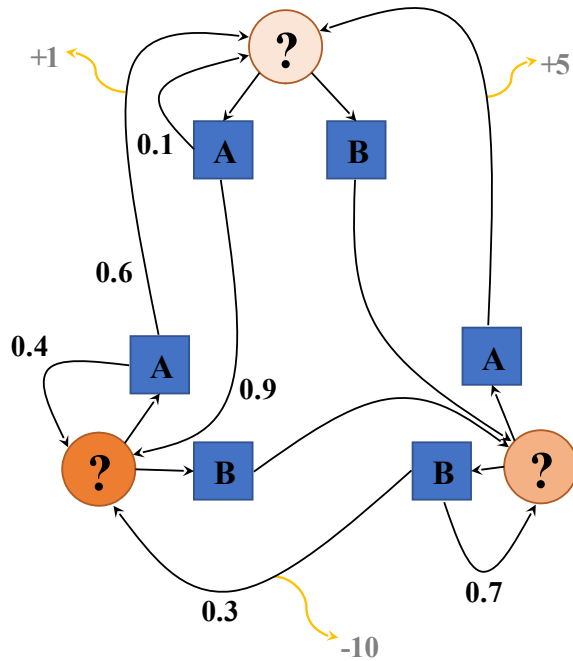


$$\mathcal{S} = 1 \quad \mathcal{O} = \{1, 2, 3\}$$

$$Z(o | s=1, a=B) = \begin{cases} 0.1 & \text{if } o=1 \\ 0.05 & \text{if } o=2 \\ 0.05 & \text{if } o=3 \end{cases}$$

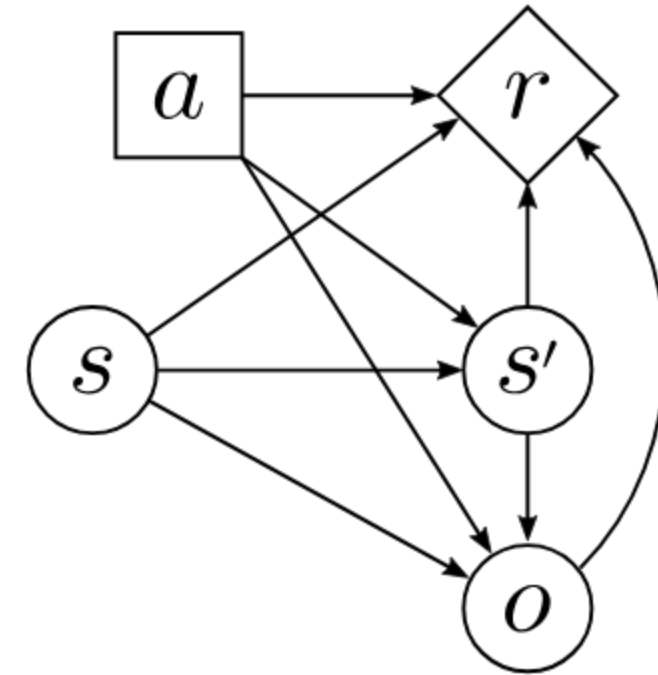
$$Z(o | s, a, s')$$

Partially Observable Markov Decision Process (POMDP)

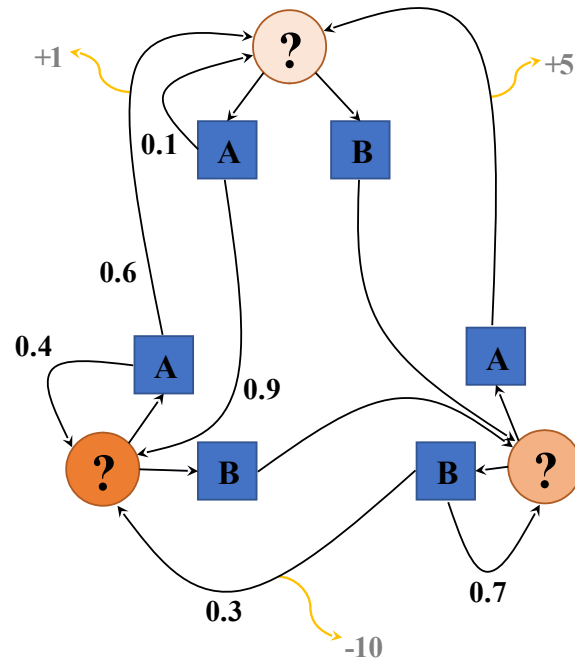


- $\mathcal{S}$ - State space
- $T(s' | s, a)$ - Transition probability distribution
- $\mathcal{A}$ - Action space
- $R(s, a)$ - Reward
- $\mathcal{O}$ - Observation space
- $Z(o | a, s')$ - Observation probability distribution

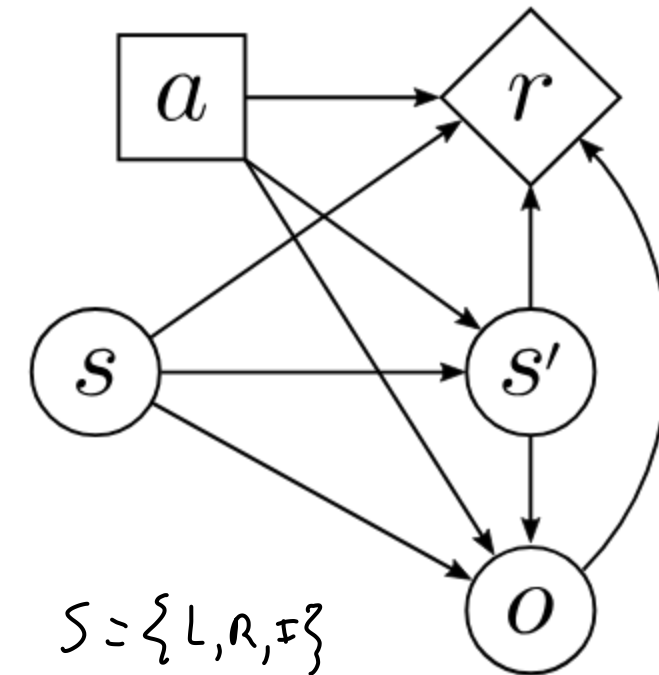
Alleatory



Partially Observable Markov Decision Process (POMDP)



- $\mathcal{S}$ - State space
- $T(s' | s, a)$ - Transition probability distribution
- $\mathcal{A}$ - Action space
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- $Z(o | a, s')$ - Observation probability distribution



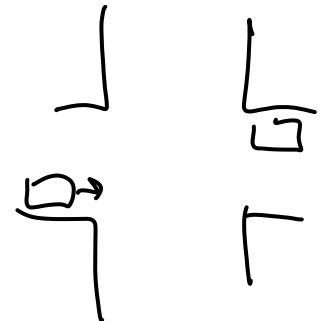
$$\mathcal{S} = \{L, R, F\}$$

$$\mathcal{O} = \{BL, BR\}$$

Alleatory

Epistemic (Static)

Epistemic (Dynamic)



Tiger POMDP Definition

SCENARIO A: TIGER IS BEHIND LEFT DOOR

STATE (S_1): TIGER LEFT



SCENARIO B: TIGER IS BEHIND RIGHT DOOR

STATE (S_2): TIGER RIGHT



Gemini

$A=L$ $A=L$ $A=L$

$O=L$ $O=L$ $O=L$

$A=L$ $A=L$ $A=L$

$O=L$ $O=R$ $O=L$

$$M = \{S, A, T, R, O, Z\}$$

$$S = \{TL, TR\}$$

$$A = \{OL, OR, Listen\}$$

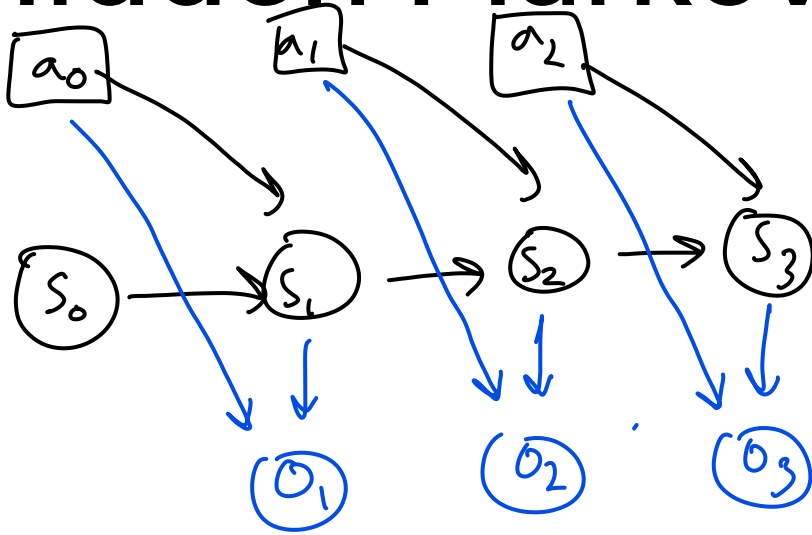
$$R = \begin{cases} -100 & \text{if } a=OL, s=TL \text{ or } a=OR, s=TR \\ 10 & \text{if } a=OR, s=TL \text{ or } a=OL, s=TR \\ -1 & \text{if } a=L \end{cases}$$

$$O = \{L, R\}$$

$$Z(o|a, s') = \begin{cases} 0.85 & o=L, s'=TL, a=Listen \\ 0.15 & o=R, s'=TL, a=Listen \\ 0.85 & o=R, s'=TR, \text{ " " } \\ 0.15 & o=L, s'=TR, \text{ " " } \\ 0 & o=L, s'=TR, a=OL \\ 1 & o=L, s'=TL, a=OL \end{cases}$$

$$T(s'|s, a) = \begin{cases} 1 & \text{if } s'=s, a \neq Listen \\ 0 & \text{o.w.} \\ 0.5 & a=OL \text{ or } OR \end{cases}$$

Hidden Markov Models and Beliefs



$$P(s_{t+1} | s_0, a_0, \dots, s_t, a_t) = T(s_{t+1} | s_t, a_t)$$

~~$$P(s_{t+1} | a_0, a_1, a_2, o_1, o_2, \dots, o_t, a_t) = P(s_{t+1} | a_t, a_t)$$~~

NOT TRUE!

$$b_0(s) = b(s)$$

↑ initial state distribution

$$h_t \equiv (b_0, a_0, o_1, \dots, a_{t-1}, o_t)$$

$$b_t(s) \equiv P(s_t = s | h_t)$$

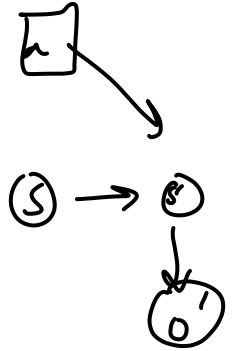
$$P(b_{t+1} | b_0, a_0, \dots, b_t, a_t) = P(b_{t+1} | b_t, a_t)$$

POMDPs are MDPs on the belief space

$$h_t = (b_0, a_0, o_1, \dots, a_{t-1}, o_t)$$

Bayesian Belief Updates

$$\begin{aligned}
 b_+ &\equiv P(s_+ | h_+) = P(s_+ | h_{t-1}, a_{t-1}, o_t) \\
 &= \underbrace{P(o_t | s_+, h_{t-1}, a_{t-1})}_{\text{observation in } s_t} \underbrace{P(s_+ | h_{t-1}, a_{t-1})}_{\text{state}} \quad \text{Bayes Rule} \\
 &\quad P(o_t | h_{t-1}, a_{t-1}) \rightarrow \text{observation across all } s
 \end{aligned}$$



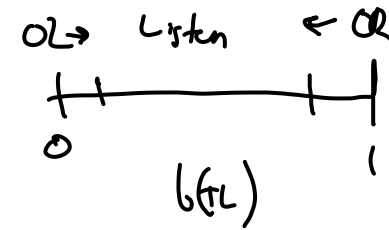
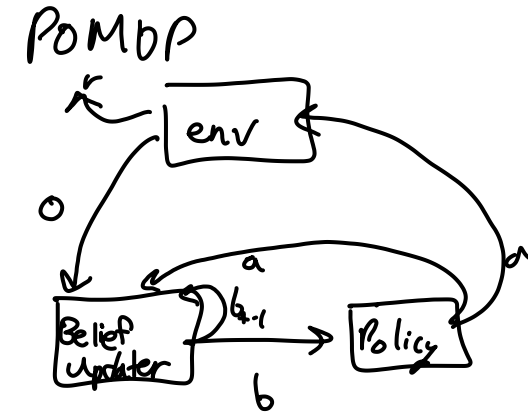
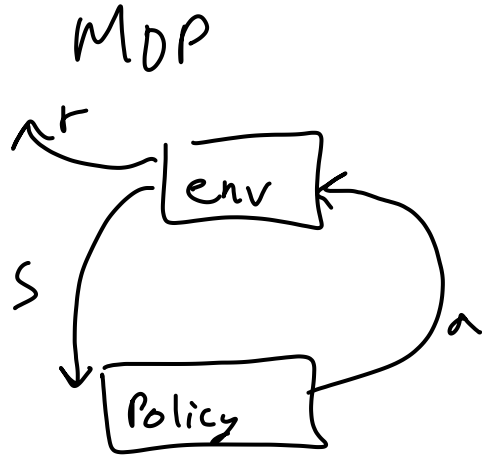
$$\begin{aligned}
 &\propto P(o_t | s_+, h_{t-1}, a_{t-1}) P(s_+ | h_{t-1}, a_{t-1}) \\
 &\quad \rightarrow \text{probability of transitioning to } s_t \quad \rightarrow \text{prob of } s_{t-1} \text{ from } h_{t-1} \\
 &= P(o_t | s_+, a_{t-1}) \sum_{s_{t-1}} P(s_+ | s_{t-1}, a_{t-1}, h_{t-1}) P(s_{t-1} | h_{t-1}, a_{t-1}) \\
 &= \text{" " " } \sum_{s_{t-1}} P(s_+ | s_{t-1}, a_{t-1}) P(s_{t-1} | h_{t-1})
 \end{aligned}$$

$$= Z(o_t | s_+, a_{t-1}) \sum_{s_{t-1}} T(s_+ | s_{t-1}, a_{t-1}) b_{t-1}(s_{t-1})$$

$$\begin{aligned}
 &\rightarrow b'(s') \propto Z(o | a, s') \sum_{s_+} \underbrace{T(s' | s_+, a)}_{\text{probability of transitioning to } s'} b(s_+) \\
 &\quad \text{probability of } o \text{ in } s' \downarrow
 \end{aligned}$$

Tiger
 $b_0 = \{0.5, 0.5\}$
 $b_1(s=TL) \quad o_1=L \quad a=Listen$
 $b(TL) \propto 0.85 \cdot (1 \cdot 0.5 + 0 \cdot 0.5)$

Filtering Loop



Tiger Example

Recap

Discussion of Types of Uncertainty

POMDP Definition $\rightarrow M = (S, A, T, R, o, z)$

Why we need beliefs

Belief Update!

